

Hardware Solutions:

Roofline : Performance

- NVIDIA GPUs
 - <https://images.nvidia.com/aem-dam/Solutions/geforce/blackwell/nvidia-rtx-blackwell-gpu-architecture.pdf>
 - <https://www.amax.com/nvidia-technical-whitepapers/>
 -
- Google TPUs
 - <https://cloud.google.com/blog/products/ai-machine-learning/an-in-depth-look-at-googles-first-tensor-processing-unit-tpu>
 - <http://tomshardware.com/news/tpu-v2-google-machine-learning.35370.html>
 - https://www.hc32.hotchips.org/assets/program/conference/day2/HotChips2020_ML_Training_Google_Norrie_Patil.v01.pdf
 - <https://blog.google/innovation-and-ai/infrastructure-and-cloud/google-cloud/ironwood-tpu-age-of-inference/>
 - Understand what's a TPU Pod
- Dmatrix
 - Corsair - <https://d-matrix.ai/pdf/d-Matrix-WhitePaper-Technical-FINAL.pdf>
 - Presentation at <https://hc2025.hotchips.org/>
 - <https://www.youtube.com/watch?v=p97UkNh2oPc&t=1767s>
 - <https://www.d-matrix.ai/going-vertical-why-we-created-a-3d-dram-solution-to-advance-low-latency-ai-inference>
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- NVIDIA + Groq LPU solution
 - <https://developer.nvidia.com/blog/inside-nvidia-groq-3-lpx-the-low-latency-inference-accelerator-for-the-nvidia-vera-rubin-platform/>
 - <https://cdn.sanity.io/files/chol0sk5/production/5c2f94426fd80fc3747c125620d9771a281a0644.pdf>
 - <https://cdn.sanity.io/files/chol0sk5/production/90632bb65692b2db9a7087b3ca6c8833f0bd24dd.pdf>
 -
- Etched Sohu
 - <https://claude.ai/chat/87028f3e-11dc-4f36-b18c-e5f69a8a1178>
 - <https://www.lesswrong.com/posts/qhpB9NjcCHjdNDsMG>
 - <https://awesomeagents.ai/hardware/etched-sohu/>
 - Roofline Model" (Williams et al.)
 - <https://techcrunch.com/2024/06/25/etched-is-building-an-ai-chip-that-only-runs-transformer-models/>
 -
- AMD Instinct / MI-300 blogs (ROCm)
 - [Deep Dive into MI300 and Memory Partitioning](#)
- Qualcomm NPUs
 - Some publicity material: <https://www.qualcomm.com/processors/hexagon>

- Chips and Cheese Blog (goes into some microarch details):
<https://chipsandcheese.com/p/qualcomms-hexagon-dsp-and-now-npu>
- <https://www.qualcomm.com/content/dam/qcomm-martech/dm-assets/documents/Unlocking-on-device-generative-AI-with-an-NPU-and-heterogeneous-computing.pdf>
- <https://www.hc2023.hotchips.org/assets/program/conference/day2/ML%20Inference/HC2023%20Qualcomm%20Hexagon%20NPU.pdf>
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- MatX
 - <https://matx.com/research> (Blogs related to their work is present in their website)
- Tenstorrent
 - [Wormhole basics](#)
 - Blackhole and Linux - [Linux on TT-Blackhole](#)
 - Open Chiplet Architecture - <https://www.openchipletatlas.org/>
- Groq LPU
 - Understanding LPU: <http://pkamath.com/publications/papers/tsp-isca20.pdf>
 - Survey Paper which has info on Groq : <https://arxiv.org/pdf/2506.00008>
 - <https://research.fpx.world/p/nvidias-christmas-eve-gift-groq-and>
 - <https://naddod.medium.com/deep-dive-into-nvidia-groq-3-lpu-a-new-choice-for-ai-inference-76eaea45bedf>
 - <https://hc34.hotchips.org/assets/program/conference/day2/Machine%20Learning/HotChips34%20-%20Groq%20-%20Abts%20-%20final.pdf> (HOT CHIPS)
 - <https://patentimages.storage.googleapis.com/4a/14/f6/007c668582be2c/US11645226.pdf> (US Patent)
 - <https://developer.nvidia.com/blog/inside-nvidia-groq-3-lpx-the-low-latency-inference-accelerator-for-the-nvidia-vera-rubin-platform/>
- Microsoft Maia
 - <https://techcommunity.microsoft.com/blog/azureinfrastructureblog/deep-dive-into-the-maia-200-architecture/4489312>
 - [https://arxiv.org/html/2601.19213v1#:~:text=MX%20formats%2C%20such%20as%20MXFP4,Xu%20and%20Ramakrishnan%2C%202024\)%20](https://arxiv.org/html/2601.19213v1#:~:text=MX%20formats%2C%20such%20as%20MXFP4,Xu%20and%20Ramakrishnan%2C%202024)%20) - Understanding Maia's compute engines by diving into Microscaling
 - <https://azure.microsoft.com/en-us/blog/azure-maia-for-the-era-of-ai-from-silicon-to-software-to-systems/> - Azure Maia intro blog
 - https://hc2024.hotchips.org/assets/program/conference/day2/81_HC2024.Microsoft.Xu.Ramakrishnan.final.v2.pdf - HOT CHIPS presentation
- Meta MTIA
 - <https://medium.com/@santhosraj14/six-ai-chips-in-two-years-inside-metas-blazing-fast-mtia-architecture-89ad165d80a4>
 - MTIA Paper found by Samarth
 - <https://ai.meta.com/blog/meta-mtia-scale-ai-chips-for-billions#:~:text=We%20achieve%20high%20velocity%20through,1%2FO%2C%20and%20networking>
 - <https://ai.meta.com/blog/meta-training-inference-accelerator-AI-MTIA/>

- Amazon Trainium/Inferentia
 - <https://awsdocs-neuron.readthedocs-hosted.com/en/latest/about-neuron/arch/neuron-hardware/trainium.html>
 - <https://newsletter.semianalysis.com/p/amazons-ai-self-sufficiency-trainium2-architecture-networking>
 - <https://aws.amazon.com/ai/machine-learning/trainium/#:~:text=Trn2%20UltraServers%20remain%20a%20high,chip%2Dto%2Dchip%20interconnect.>
- Taalas
 - <https://medium.com/@mokrasar/the-last-chip-how-hardwired-ai-will-destroy-nvidias-empire-and-change-the-world-8da20571e706>
 - <https://o-mega.ai/articles/taalas-guide-the-17-000-tps-ai-chip-revolution-2026>
 - https://www.reddit.com/r/singularity/comments/1r9frzk/taalas_llms_baked_into_hardware_no_hbm_weights/#:~:text=They%20have%20a%20very%20radical,onto%20a%20single%20silicon%20chip. (Reddit thread lol , but some insights are there)
- Cerebras
 -  Cerebras @ Hot Chips 34 - Sean Lie's talk, "Cerebras Architecture Deep D..."